

AI in Diabetes Care: Practical Applications for Healthcare Professionals

Introduction

It's a pleasure to welcome you to this course on artificial intelligence in diabetes care. AI is rapidly finding its way into everyday diabetes tools: from CGM apps that forecast glucose trends, to software that suggests insulin adjustments, triages retinal images, or drafts clinical notes. These systems promise better prediction, less administrative burden, and more personalized care—but only if we understand what they are actually doing.

However, it is crucial to remember that there are important caveats.

- Not all “AI-enabled” tools are the same; some are simple rule-based systems, others are complex data-driven models, and many are hybrids.
- It is vital to know which data a model learns from and in which patient population it has been validated.
- We need to recognize where AI can safely support decisions, and where human clinical judgement must clearly remain in the lead.

These are precisely the topics this master course on AI in diabetes care will cover. In the upcoming lessons, you will explore:

- A practical framework for understanding AI, including the difference between rule-based and data-driven systems, and how terms like machine learning, deep learning, generative AI, and large language models fit together.
- How large language models can support people with diabetes and healthcare professionals, for example by explaining concepts, summarizing reports, and preparing patient education—together with clear boundaries on what they must not do, such as insulin dosing.
- AI in glucose prediction, from familiar hypoglycaemia alerts to full glucose-curve forecasting and digital twins, and how to interpret performance metrics in a clinically meaningful way.
- AI in insulin-dosing decision software, where you will see how open-loop and closed-loop systems work today, where genuine machine learning is already used, and why safety layers and human oversight are non-negotiable.
- AI in carbohydrate estimation and carbohydrate counting, focusing on where AI tools can meaningfully support carb estimation, where current systems remain fragile—especially for mixed meals and under-represented

cuisines—and why rare extreme errors make them unsuitable as the sole basis for insulin bolus calculations.

- AI in complication screening, focusing on current applications such as retinopathy screening and emerging tools for kidney disease, neuropathy, the diabetic foot, and heart failure.
- Ethical, legal, and privacy aspects, including bias and equity, GDPR and data governance, the EU AI Act, and your non-delegable responsibilities as a clinician when AI is part of care.

With this knowledge, you will be able to look at any AI claim in diabetes care and ask the right questions: What type of system is this? What data does it rely on? How was it evaluated? How much autonomy does it have? And, most importantly, does it actually improve care for my patients?

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All material is completely independent from device manufacturers and is intended as education, not as individual medical advice; for any changes in therapy, always consult the responsible healthcare team and follow local guidelines and regulations.

Lesson 1: What Is AI? A Framework for Clinicians

Welcome to this course on artificial intelligence in diabetes care.

Over the next eight lessons, we'll explore how AI is changing the way we predict glucose levels, dose insulin, screen for complications, and manage our daily clinical workload.

But before we get into the applications, we need to build a shared framework. Because when someone says "this device uses AI" — you actually know almost nothing yet. You need to know which type of AI, what data it uses, what task it performs, and how much it can be trusted.

That is exactly what this first lesson is about.

By the end, you'll have a practical mental map that you can apply to any AI claim you encounter — in a conference presentation, a product brochure, or a patient asking about their device.

AI is not 1 thing.

The first thing to understand is that AI is an umbrella term. It covers a very wide range of approaches — from simple rule-based systems to complex neural networks.

Look at the diagram on the right. You can see that machine learning is a subset of AI. Deep learning is a subset of machine learning. Generative AI sits within deep learning. And large language models, or LLMs, are a specific type of generative AI. Natural language processing and computer vision are separate branches that overlap with several of these categories.

This nesting is important, because it means these terms are not alternatives to each other — they are layers of description. A system can be machine learning, deep learning, and generative AI all at once.

So the first question to ask is always: what kind of AI are we actually talking about?

Rule-based versus data-driven

The most clinically important distinction is not between AI and "non-AI." It's between systems that follow explicit rules and systems that learn from data.

Rule-based systems follow logic that humans have programmed in advance. A good example is the predictive hypoglycaemia alert in most CGMs. The system calculates: if glucose is dropping at this rate, it will cross the threshold in twenty minutes — trigger an alert. The rules are clear, the output is predictable, and you can understand exactly why the alarm fired.

Data-driven systems — also called machine learning — learn patterns from large datasets without being told the rules explicitly. Instead of a clinician writing "if glucose drops by 2 mg/dl per minute, then alert," the system analyses thousands of CGM traces and discovers on its own which patterns predict hypoglycaemia.

As you can see in the figure, moving from rule-based to more complex AI models typically increases prediction accuracy — but reduces how easy the model is to interpret. A random forest is harder to explain than a decision tree. A deep neural network is harder to explain than a random forest. This trade-off between accuracy and transparency is something you will encounter repeatedly in clinical AI.

Both approaches have value. Rule-based systems are transparent and easy to validate. Data-driven systems can find complex patterns that no human would have programmed. Many real-world systems combine both — and that is particularly true for closed-loop insulin delivery systems.

Closed-loop debate

This is exactly why the question "are closed-loop systems AI?" has no simple yes or no answer. As you can see in the figure, current closed-loop systems sit at the intersection of multiple approaches — control engineering, rule-based algorithms, Bayesian methods, and in some research systems, machine learning. The answer depends entirely on which definition of AI you are using.

Three types of machine learning

Within machine learning, there are three main ways a system can learn.

Supervised learning means the system learns from labelled examples. For retinopathy screening, you show the model thousands of fundus photos, each labelled as "retinopathy present" or "not present," and it learns which visual patterns matter. For glucose forecasting, you show it CGM traces paired with the actual glucose values that followed, and it learns to predict ahead.

Unsupervised learning means the system finds structure without being told what to look for. In diabetes research, this is used to discover patient subgroups — for example identifying clusters of type 2 diabetes patients with similar glucose profiles, which might respond differently to treatment.

Reinforcement learning means the system learns through trial and feedback. An algorithm tries different insulin doses, receives a signal about whether glucose stayed in range, and gradually learns a better strategy. This approach is being explored in experimental closed-loop systems.

Deep learning and neural network architectures

An important clarification before we continue: deep learning is not a fourth type of machine learning. Deep learning refers to using neural networks with many layers — and this architecture can be applied within any of the three learning types above. There are several types of neural network architectures, and they are suited to different kinds of data.

Convolutional neural networks, or CNNs, are designed for images. They are the foundation of AI-powered retinopathy screening and meal photo recognition.

Recurrent neural networks, including LSTMs, are designed for sequential data over time. A CGM glucose trace is a perfect example — the model needs to "remember" what happened in the past 30 or 60 minutes to predict what comes next. This is how the Accu-Chek SmartGuide Glucose Predict feature works: a recurrent neural network trained on three large CGM datasets, producing a two-hour glucose forecast.

Transformers are the architecture behind large language models. They use a mechanism called attention to understand relationships between words or data points, regardless of how far apart they are in a sequence.

The key message here is that architecture — how the model is built — is a separate axis from learning method — how the model is trained. Knowing both gives you a much more complete picture of what a system actually does.

What the model does - its function

Alongside how a model learns, you also need to know what it does.

Predictive AI classifies or forecasts — hypoglycaemia risk, glucose trajectory, retinopathy grade.

Generative AI creates new content — text, summaries, images. Large language models like ChatGPT are the most familiar example. These are relevant in diabetes care for documentation, patient communication, and clinical decision support, which we'll cover in Lesson 2.

Control systems actively steer a process — like closed-loop insulin delivery. These are not the same as predictive AI, and not the same as generative AI. They sit in their own category.

Agentic AI - the next step

You may be hearing more and more about "AI agents" or "agentic AI." This is an important new concept.

A regular AI tool responds to one question at a time. You ask something, it answers, and it stops.

An AI agent is different. You give it a goal, and it plans and executes a series of steps to reach that goal — autonomously. It can retrieve data, consult guidelines, summarise findings, and deliver a result — all without you having to guide it step by step.

In diabetes care, you could imagine an agent that, given a patient's name before a consultation, automatically retrieves the latest CGM data, checks recent lab results, identifies relevant guideline recommendations, and prepares a structured consultation summary — all before you walk into the room.

This is not yet standard practice in diabetes clinics. But it is the direction in which clinical AI is moving. And it raises important questions about oversight, accountability, and when it is safe to let an AI system act without a human reviewing each step.

Agentic AI is not a separate learning method — it is an autonomy layer. An agent can use any combination of AI tools internally: an LLM for language, a predictive model for risk, a retrieval system for guidelines. What makes it agentic is that it takes initiative across multiple steps toward a goal.

How autonomous is the system?

One more axis matters clinically: how much does the system act on its own?

At one end, AI simply displays information — a CGM pattern summary, a risk score. A clinician or patient still makes all decisions.

In the middle, AI recommends — a bolus suggestion, a therapy change recommendation. A human reviews and accepts or rejects it.

At the far end, AI acts autonomously — a closed-loop system adjusting basal insulin every five minutes without asking permission.

This matters because the level of autonomy determines the regulatory requirements, the validation standards, and the clinical responsibility that falls on you as a prescribing clinician. An AI scribe that makes an error costs you time. An AID system that makes an error can harm your patient.

Locked versus adaptive models

One final distinction worth knowing: does the model change after it has been validated?

A locked model is fixed after approval. It behaves the same way every time, which makes it easier to test and regulate. Most commercial diabetes devices use locked models.

An adaptive model continues to learn or adjust during use. CamAPS FX, for example, uses Bayesian adaptation to personalise its algorithm to each individual over time. This can improve performance — but it also makes validation more complex, because the system you validated at launch is not exactly the same system your patient is using six months later.

The ANI to AGI to ASI spectrum

You will sometimes hear terms like artificial general intelligence, or AGI, and artificial superintelligence in media discussions about AI.

It is worth knowing where we actually are on this spectrum.

We are firmly in the era of Artificial Narrow Intelligence, or ANI. Every AI system you will encounter in clinical practice today — including the most sophisticated closed-loop algorithm or the most capable LLM — is narrow AI. It is very good at one specific type of task, and it cannot transfer that ability to unrelated tasks.

AGI — a system as capable as a human across all cognitive domains — does not exist yet. ASI — a system that surpasses human intelligence across all domains — is speculative. These are important concepts for following the public debate about AI, but they are not current clinical reality.

In clinical practice, the relevant question is not "how close is this to AGI?" — it is "what specific task does this narrow AI perform, and how well does it perform it in my patient population?"

A practical checklist

Whenever you encounter an AI claim, ask these questions:

1. Is this rule-based, data-driven, or a hybrid?
2. What data does it learn from — and in which patient population?
3. What does it output — a prediction, a recommendation, or a direct action?
4. How was it evaluated — sensitivity and specificity for binary outcomes; RMSE or Clarke Error Grid for continuous glucose predictions?
5. How autonomous is it — does a human stay in the loop?
6. Is it locked after validation, or does it keep learning after deployment?

You don't need to be an AI engineer to ask these questions. But asking them will help you evaluate any system with the same rigour you'd apply to a clinical trial.

In the lessons that follow, we'll apply this framework to concrete diabetes applications.

Lesson 2 covers large language models — what they can and can't do in clinical practice, and how to use them responsibly. Lesson 3 focuses on AI in glucose forecasting, including the specific algorithms behind predictive alerts and glucose curve prediction. Lesson 4 covers AI in insulin dosing, from bolus calculators to fully closed-loop systems. Lesson 5 looks at AI in carbohydrate estimation and carbohydrate counting — where current tools can genuinely support carb estimation, where they fail (especially for mixed meals and under-represented cuisines), and why rare extreme errors make them unsafe as the sole basis for bolus calculations. Lesson 6 covers AI in complication screening, particularly retinopathy and foot assessment. And Lesson 7 addresses the ethics, governance, and safe implementation of AI in your clinical setting.

Each lesson is designed to be practical. The goal is not to make you an AI expert — it is to make sure you can use, evaluate, and advocate for AI tools safely and confidently.

Let's get started.

Lesson 2: LLMs for Healthcare Providers and People With Diabetes

Welcome. In this lesson, we will talk about large language models, or LLMs, for healthcare providers and people with diabetes.

The most important message is this: LLMs can be very useful for understanding information, preparing questions, summarizing data, translating instructions, and creating educational material. But they are not clinicians, they are not regulated insulin-dosing tools, and they should not be used as autonomous medical decision-makers. The safest clinical framing is: use LLMs as a second opinion, not as a first authority.

This matters because people are already using these tools. OpenAI has reported that more than 230 million people ask health and wellness questions on ChatGPT each week, and Reuters reported that OpenAI's health feature was designed partly because health questions had become a major use case [1]. Physicians are using LLMs as well: one 2025 survey found that 64.4% of respondents reported using LLMs, while only 2.6% recommended them to patients [2].

So the question is not whether patients and clinicians will use LLMs. They already do. The question is how to use them safely.

Which LLMs are people using?

In daily practice, the best-known general models are ChatGPT, Claude, and Gemini. These models change quickly, so it is useful to look at live ranking pages rather than memorize a fixed order. OpenRouter provides a general LLM leaderboard using benchmarks, weekly usage, trending status, and real-world usage data [3]. MedArena is more medically focused and compares medical LLMs using human preference data, pairwise win rates, Bradley-Terry ratings, Elo-style scores, confidence intervals, and labels for features such as RAG and vision support [4].

But a leaderboard is not a license to use a model for clinical decisions. A model can score well on a benchmark and still fail on a specific patient, a specific device, a local guideline, or a safety-critical instruction.

What is an LLM actually doing?

An LLM is not a database, not a doctor, and not a calculator by default. It is a language model trained to predict and generate plausible language.

Its pretrained knowledge reflects patterns in training data. Common topics are often well represented, while rare topics, proprietary device details, local hospital protocols, or recent guideline updates may be missing or poorly represented. Training data can also contain errors from forums, videos, Wikipedia pages, news articles, and even older academic content. The model does not automatically know which source is best for your clinical purpose.

Another important concept is the distinction between the model's internal knowledge and external information retrieval. The pretrained knowledge of an LLM is frozen at a certain point in time. However, many modern AI systems can now access the web, search engines, or connected databases to retrieve more recent information. When these search capabilities are enabled, the system can often provide up-to-date information beyond its original training data.

For medical work, this distinction matters. Some questions can be answered from the model's existing knowledge, while others require current information from guidelines, regulatory agencies, manufacturers, or newly published research. If you need device manuals, peer-reviewed evidence, professional guidelines, or regulatory documents, ask for them explicitly. Rather than asking, “What is the answer?”, ask: “Search peer-reviewed sources, guidelines, and manufacturer documents, and cite them.”

What the evidence tells us

The evidence is developing quickly, but the pattern is consistent: LLMs are often strong at language-heavy tasks, but weaker when the task becomes individualized, numerical, guideline-dependent, or safety-critical.

In diagnostic reasoning, a JAMA Network Open randomized trial found that access to GPT-4 did not significantly improve physicians' diagnostic reasoning scores compared with conventional resources, even though GPT-4 alone scored highly on the same case set [5]. A practical interpretation is that the model is not enough by itself: clinicians need the right workflow, context, and training.

In diabetes education, structured prompting can improve answers. The RISE framework improved model answers to diabetes questions, with GPT-4 improving

from 91% to 98%, Claude from 72% to 91%, and Bard from 86% to 95% in that study's question set [6]. This supports a practical lesson: better structure gives better answers.

In CGM interpretation, one Scientific Reports case study found that GPT-4 correctly calculated 9 of 10 quantitative CGM metrics across 10 cases, but was less reliable for one time-above-range metric [7]. The message is: impressive performance still needs checking.

For carbohydrate estimation from meal images, performance is much better for simple foods than for complex mixed meals. One study found high agreement for fruits and vegetables, but much lower agreement for composite meals [8]. Another study of 246 hospital meals found mean absolute carbohydrate errors ranging from approximately 13 grams to more than 20 grams depending on the tool [9].

This leads to a critical safety principle: fluency is not accuracy. An LLM can describe an incorrect insulin dose, an incorrect carbohydrate estimate, or an incomplete safety plan in the same confident language it uses when it is correct.

What LLMs can do well in diabetes care

LLMs can explain: for example, "Explain time in range in simple language," or "What does insulin on board mean?" They can summarize long notes, device reports, meeting transcripts, or patient instructions. They can translate and adapt content into another language, a lower reading level, or a checklist. They can also help a patient prepare for a visit by generating questions to discuss with the diabetes team.

LLMs can work with uploaded documents. If you upload a pump manual, local protocol, guideline, or paper, the model can answer based on those documents. This is close to retrieval-augmented generation, or RAG.

What LLMs should not do

LLMs should not autonomously prescribe insulin, change pump settings, adjust basal rates, change correction factors, or tell a patient to start, stop, or change medication.

The Diabetes Technology Network UK states that generic LLMs can be supplementary tools for tasks like initial carbohydrate estimation from food images or identifying glucose patterns, but they are not a substitute for medical expertise and cannot safely provide definitive medical advice or insulin dosing recommendations [10].

LLMs should also not be trusted blindly for carbohydrate counting. In insulin-treated diabetes, the average error is not the only issue. A rare large overestimate or underestimate can matter clinically. LLMs also tend to be agreeable: if you ask a biased question, you may get a biased answer.

So in medicine, prompt neutrally. Instead of asking, “Why is this pump setting wrong?”, ask: “Review this de-identified situation. List arguments for and against changing the plan. State what information is missing. Do not make a dosing recommendation.”

Patient data and privacy

There is one safety boundary that must be explicit: do not paste identifiable patient information into open consumer LLMs.

That means no names, birth dates, national numbers, contact details, identifiable screenshots, unredacted clinic letters, or anything that could reasonably identify a patient. For clinicians in Europe, this is also a GDPR issue.

There are exceptions. Some clinicians may have access to closed enterprise LLMs, hospital-approved systems, locally deployed models, or platforms with a signed data processing agreement. In those settings, patient data may sometimes be used, but only according to local policy and institutional approval.

As a default: anonymize or pseudonymize first. If the task can be done without patient identifiers, remove them.

How to use LLMs well

First, give context. For example: “I am a diabetologist preparing patient education for adults with type 1 diabetes starting hybrid closed loop. Use plain English. Avoid dosing advice. Include safety warnings.”

Second, upload trusted sources when appropriate: guidelines, device manuals, patient handouts, journal articles, or local protocols. Then say: “Use the uploaded document as the primary source. If you use outside information, cite it separately.”

Third, ask for the format you want: a checklist, patient handout, clinician summary, FAQ, troubleshooting guide, script, email, or slide outline.

Fourth, ask for uncertainty: “Tell me what you are confident about, what is uncertain, what information is missing, and what should be checked with a clinician.”

Fifth, ask for alternatives. Ask for three versions, then give feedback. You can also ask the model to improve your prompt before answering.

Finally, use voice input if it helps. Tools such as Wispr Flow and other modern dictation apps can make prompting faster. This matters because more context often leads to better answers.

RAG in simple terms

RAG means retrieval-augmented generation.

In simple terms, the model first retrieves relevant information from a trusted source, then uses that information to generate an answer. When you upload documents into ChatGPT, Gemini, or Claude and ask the model to base its answer on them, you are using a simple form of RAG. More advanced RAG systems can be built into custom GPTs, Gemini Gems, Claude Projects, or institutional tools.

The key is source quality. If the system searches random internet content, it may retrieve random internet quality. If it searches peer-reviewed journals, professional guidelines, regulatory documents, and validated local protocols, the answer is more likely to be useful.

However, RAG is not magic. The quality of the answer depends not only on the quality of the source material but also on whether the system retrieves the correct information. RAG systems can retrieve the wrong paragraph, miss important context, overlook exclusion criteria, or fail to identify the most relevant section of a large document. This becomes increasingly challenging when working with lengthy guidelines, device manuals, or complex medical documents.

Even when the correct information is retrieved, the model may still misinterpret it, combine it incorrectly with other information, or present it in a misleading way. For this reason, RAG can improve accuracy but does not guarantee correctness. Clinical review and professional judgment remain essential.

Practical prompts

For a person with diabetes:

“I use a CGM and I see high glucose after breakfast. I do not want insulin dosing advice. Help me understand possible reasons, such as meal composition, timing, infusion set issues, stress, or activity. Give me questions to discuss with my diabetes team.”

For a meal image:

“Estimate the carbohydrates in this meal image as a rough educational exercise only. Give a range, explain uncertainty, and remind me that I should use labels, weighing, or my diabetes team’s instructions for insulin decisions.”

For a clinician:

“Summarize this de-identified CGM report into a patient-friendly paragraph. Do not recommend dose changes. Focus on patterns, questions for discussion, and safety.”

For privacy:

“Before I paste anything, list what patient identifiers I should remove so that this case is de-identified.”

Closing

The safest way to think about LLMs in diabetes is this: they are powerful language and reasoning assistants, not autonomous clinicians.

They can help people with diabetes understand devices, meals, instructions, and troubleshooting steps. They can help clinicians prepare summaries, translations, educational material, and structured explanations. But for insulin, medication changes, pump settings, acute illness, pregnancy, severe hypoglycemia, ketones, or any urgent situation, the boundary is clear: the LLM can help prepare questions and explain general concepts, but it should not replace clinical care.

Use LLMs with context. Use trusted sources. Ask for uncertainty. Ask for citations. Protect patient data. Keep the human in the loop. That is how we move from “AI gives an answer” to “AI supports safer diabetes communication.”

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Lesson 3: AI in Glucose Prediction

Welcome to lesson 3 on AI in glucose prediction. The key message of this lesson is that glucose forecasting is not one single concept. It can mean predicting hypoglycemia, predicting a future glucose trace, or predicting what may happen after a treatment change. If diabetologists understand those different use cases, they can better judge where AI is already useful and how to evaluate whether a system is clinically meaningful.

The first and most familiar form of glucose prediction is hypoglycemia prediction. In fact, many clinicians are already using predictive glucose technology every day through CGM alerts. Several commercial CGM systems provide alerts that estimate whether glucose is likely to cross a low threshold within a certain time window, such as 20, 30, or 60 minutes. These alerts are typically based on deterministic logic, meaning that the system uses the current glucose and the rate of change to estimate where glucose is heading next.

So even before we talk about machine learning, we should recognize that predictive alerts are already a form of algorithmic glucose forecasting. For example: if glucose is falling at about 1 milligram per deciliter per minute, then over 30 minutes the projected drop is about 30 milligrams per deciliter. That is not machine learning, but it is still a useful predictive framework that feels familiar and clinically intuitive.

What is new is that machine learning is now being added to this same clinical problem. A clear commercial example is the Accu-Chek SmartGuide system, which incorporates three AI-enabled predictive features: Low Glucose Predict, Night Low Predict, and Glucose Predict, helping users anticipate and proactively manage future glucose trends.

The SmartGuide Low Glucose Predict feature estimates the probability of a low-glucose event occurring within the next 30 minutes using an XGBoost machine-learning model. The model is based on five input features derived from the current CGM value, glucose trends over different time windows, logged carbohydrates and time-related information.

The system also includes a Night Low Predict feature, which estimates the probability of glucose falling below 70 mg/dL (3.9 mmol/L) overnight. This prediction is generated using an XGBoost model that incorporates current and historical CGM values, glucose trends, past nocturnal hypoglycemic events, previous nocturnal hypoglycemia risk, time of day, and bolus insulin on board.

Both hypoglycemia prediction algorithms use XGBoost models that were developed and trained using data from the proprietary PREDICT study, which included adults with type 1 diabetes treated with multiple daily injections. The models were subsequently evaluated on people with type 1 diabetes on MDI in the PREDICT study who were not part of the training set, and additional independent datasets, including the REPLACE-BG study in people with type 1 diabetes using insulin pumps and a real-world mySugr dataset of people with type 2 diabetes, demonstrating consistent performance across different populations and treatment modalities. [1]

This leads to the first major evaluation question for any AI-enabled glucose prediction tool: what data was the system trained and validated on? That matters because performance in one dataset does not automatically mean performance in your own patient population. If the training and validation data are narrow, highly selected, or not representative of your patients, then the algorithm may not generalize well in routine practice.

The second evaluation question is accuracy, but accuracy depends on the type of prediction. In hypoglycemia prediction, the outcome is essentially binary: yes or no, low glucose is expected or not. For that kind of task, the most relevant metrics are sensitivity, specificity, and the ROC area under the curve. For example, Roche's Accu-Chek SmartGuide Low Glucose Predict feature achieved a sensitivity of approximately 95%, a specificity of approximately 99%, and a ROC AUC of 0.96 in validation studies. [1] The Night Low Predict feature achieved a lower sensitivity of approximately 54%, but a specificity of approximately 92% and a ROC AUC of 0.86.

[1] In practice, this means the overnight model detected over half of the nights that

would be followed by hypoglycemia, but it was very good at recognizing nights that were unlikely to result in hypoglycemia. As a result, users can have relatively high confidence when the system predicts a low risk, although some hypoglycemic nights may still be missed.

The second major meaning of glucose forecasting is predicting a whole glucose trace rather than a simple yes-or-no event. This is already done in automated insulin delivery systems, but mostly with deterministic algorithms such as model predictive control, proportional-integral-derivative control, or fuzzy logic.

However, machine learning is now also being used for glucose trace forecasting. In the Accu-Chek SmartGuide system, the Glucose Predict feature estimates a 2-hour glucose forecast. It uses a recurrent neural network trained on data from 200 adults with type 1 diabetes in the PREDICT study and subsequently validated in people with type 1 diabetes on MDI in the PREDICT study who were not included in the training set and in independent datasets including people with type 1 diabetes using insulin pumps (from the REPLACE-BG dataset) and people with type 2 diabetes using the mySugr app, demonstrating consistent performance across different populations.

The algorithm uses current and past CGM values, bolus insulin on board, unabsorbed carbohydrates, and time of day to generate a two-hour glucose forecast. [1] This is an important shift because the model is not just projecting a line from current glucose; it is integrating more context about what may happen next. Notably, all three prediction features remain functional even in the absence of meal and insulin logging, enhancing their applicability in routine clinical practice where data entry is often incomplete or inconsistent.

When evaluating this type of continuous prediction, metrics such as sensitivity, specificity, and ROC AUC are no longer appropriate because the outcome is not binary. Instead, performance is assessed using continuous error metrics such as the root mean square error (RMSE) and the Consensus Error Grid (CEG). Lower RMSE values indicate that predicted glucose levels remain closer to the actual glucose values, while a higher percentage of predictions in Consensus Error Grid zones A and B indicates greater clinical accuracy.

On a pooled validation dataset, the Glucose Predict model achieved 99.8%, 99.3%, 98.7%, and 96.3% of predictions within Consensus Error Grid zones A and B at prediction horizons of 30, 45, 60, and 120 minutes, respectively. [1] These results

suggest that the model maintains a high degree of clinical accuracy, even when forecasting glucose levels up to two hours into the future.

At the research level, the field is also moving toward foundational models. TimeGPT and GluFormer are examples of this next wave. [2, 3] These models are exciting because they may generalize across more diverse glucose patterns and may eventually do more than short-term forecasting alone. But for now, the safest message is that they are promising, not yet the new clinical standard.

And that brings us to the most important clinical question: what is the actual impact of glucose prediction? A model can be accurate without improving patient outcomes. To demonstrate real value, prediction usually needs to be linked to an action, such as an insulin dose adjustment, a behavior recommendation, or a safety intervention. Without that extra layer, glucose forecasting remains informative, but its clinical effect is hard to interpret.

That is also why glucose forecasting connects directly to the next lesson on AI and insulin dosing. Prediction alone does not lower HbA1c or prevent hypoglycemia; what matters is what the person or the system does with the prediction. So the full clinical benefit often depends on an additional decision-support module sitting on top of the forecasting layer.

A third use of AI in this area is not simply predicting the next glucose value, but predicting what might happen after changing treatment. This is where digital twins come in. Digital twins are models that can replay or simulate what might happen after a change in meals, insulin, exercise, or therapy settings. They are especially valuable because they make treatment consequences visible before applying them in real life. [4]

Examples include ReplayBG, the DailyDose concept, and the ABC study. [5, 6, 7] In these applications, people with type 1 diabetes can access their digital twin and explore what-if treatment scenarios, before changing their insulin dosing settings. While digital twins are still mainly research tools and need more validation and regulatory development before broader clinical use, this is a good example of how AI can support understanding and decision-making, even before it becomes fully automated care.

So where are we today? In current clinical practice, machine learning is not yet the dominant approach in glucose forecasting. Most real-world systems, especially closed-loop systems, still rely primarily on deterministic algorithms. But machine-learning use cases are clearly emerging, and SmartGuide is a concrete example where commercial glucose prediction already incorporates models such as XGBoost and recurrent neural networks.

The practical takeaway for diabetologists is therefore quite simple. First, understand what type of glucose prediction a system is offering: binary event prediction, continuous glucose forecasting, or treatment simulation. Second, ask on what data it was trained and validated. Third, check whether the reported metric matches the task: sensitivity, specificity, and ROC AUC for hypoglycemia prediction, and RMSE for continuous glucose forecasting.

And finally, always ask the clinical question behind the technical one. Even a technically elegant model is only useful if it improves glucose control, reduces burden, supports safer treatment decisions, or helps patients and clinicians act earlier and better. That is the lens through which AI in glucose prediction should ultimately be judged.

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Lesson 4: AI in Insulin-Dosing Decision Software

Welcome to this lesson on AI in insulin dosing.

In the next ten minutes, the focus will be on where artificial intelligence already plays a role today, where systems are still driven by classical, deterministic algorithms, and why this distinction matters for safety and clinical governance.

To begin, it is useful to distinguish clearly between open-loop and closed-loop or automated insulin decision support systems.

- Open-loop systems help to calculate or recommend insulin doses, but the person with diabetes – or the clinician – remains in charge of actually delivering that insulin.
- Closed-loop systems, by contrast, not only compute the dose but also automatically command the pump, usually every few minutes, based on continuous glucose monitor data.

In other words, open loop supports human decisions; closed loop is the controller.

Historically, open-loop tools were almost entirely rule-based. Bolus calculators on pumps and in smart pens used fixed formulas based on carbohydrates, insulin-to-carbohydrate ratio, insulin sensitivity factor, current glucose, and insulin

on board. Basal titration algorithms for type 2 diabetes adjusted once-daily insulin according to fasting glucose, using deterministic, transparent rules. These approaches are still widely used and remain an important foundation of insulin therapy.

More recently, AI-enabled open-loop decision support systems have appeared. One example is the DailyDose approach, which uses a recommender system based on k-nearest neighbours to suggest adjustments to short- and long-acting insulin settings. Early studies showed only modest improvements in time in range, similar to other systems using approaches such as reinforcement learning or case-based reasoning. [1]

Newer experimental systems combine digital twins with neural networks for glucose prediction and insulin recommendations. However, most results still come from simulator studies, with limited long-term human data available so far. [2]

When moving from open loop to closed-loop automated insulin delivery, most commercial automated insulin delivery systems today still rely on deterministic, rule-based algorithms. They use proportional–integral–derivative control, model predictive control, or fuzzy logic rule sets that compute insulin delivery from a mathematical model of glucose–insulin dynamics. For a given state – defined by sensor glucose, insulin on board, and sometimes carbohydrates on board – the output is always the same. This determinism is a core safety property: it allows formal verification and exhaustive testing of possible states.

A key example of genuine AI entering closed loop is AIDANET, developed by the University of Virginia.

AIDANET is architecturally very different from classical model predictive controllers. It consists of several coordinated components. A neural-network-based basal controller receives an estimate of the current physiological state – including recent CGM data, insulin on board, and the presence of meals or exercise – and outputs a basal delivery rate. This network has been trained extensively in silico on large cohorts of virtual patients, learning a mapping from state to insulin action that is no longer fully interpretable as a set of human-readable equations.

On top of the basal controller, AIDANET includes an automated meal component. The system monitors CGM signals for patterns consistent with meals and can

deliver priming boluses before the full post-meal spike is visible, without explicit carbohydrate counting or meal announcement. The most novel layer uses a transformer network together with reinforcement learning for meal bolus sizing. A transformer processes around two weeks of an individual's CGM and insulin history and compresses this into a representation of that person's usual meal responses, insulin sensitivity and time-of-day patterns. A reinforcement-learning agent then takes this representation and outputs a meal bolus as a fraction of total daily insulin, bypassing fixed insulin-to-carb ratios or explicit correction factors.

In clinical trials, AIDANET has shown that fully closed-loop insulin-only control, without manual meal boluses, can achieve time in range comparable to modern hybrid closed-loop systems, while reducing the number of user-initiated boluses to almost zero. [3] This suggests that AI-based controllers can take over more of the day-to-day dosing burden without worsening overall outcomes, at least in supervised study settings. At the same time, such systems require strong safety layers – including hard caps on per-bolus insulin and hypoglycaemia protection – to make the behaviour of a learned controller acceptable for human use.

Between deterministic commercial AID and fully AI-based controllers like AIDANET lies an important intermediate space where machine learning augments closed-loop systems without replacing the entire controller. One example is the AIMI model on AndroidAPS, an open-source environment that allows researchers to integrate experimental machine-learning components – for instance, context-aware adjustments based on sleep, activity or hormonal cycles – on top of a well-understood rule-based algorithm. In such hybrid setups, the core insulin computation remains transparent and deterministic, while ML modules add personalisation and context sensitivity.

Another rapidly evolving area is AI-driven meal detection within automated insulin delivery. Traditional hybrid closed-loop systems are built on the assumption that users announce meals and count carbohydrates. In real life, however, meals are often under-estimated or not announced at all. Machine-learning-based meal detection tries to bridge this gap by continuously analysing CGM curves for signatures of food intake – such as the rate and acceleration of glucose rise, time of day, and recent insulin on board – and then switching the controller into a more aggressive mode when a meal pattern is detected.

The new Vivera algorithm represents this type of approach in a commercial system. It is designed to handle both announced and unannounced meals seamlessly. When the model is confident that a meal has occurred, it can deliver automatic meal boluses or temporarily lower internal glucose targets, even in the absence of a manual bolus command. For users, this means the pump may visibly “take over” more of the post-meal control. For clinicians and educators, it introduces new failure modes: during exercise, illness, or sensor artefacts, the CGM pattern can resemble a meal, leading to unnecessary aggressive dosing if the detection model is not calibrated carefully.

Because insulin is a high-risk medication, there is growing interest in safety architectures that allow innovation at the algorithm level without compromising basic protection. An elegant example is the GlucOS concept. In this design, every insulin command from the controller – whether it comes from a classical model predictive control, a neural network, or a reinforcement-learning agent – must pass through a small, deterministic “safety kernel”. This kernel enforces hard physiological constraints, such as maximum safe basal rates, limits on per-bolus insulin, and specific responses after hypoglycaemia. The safety kernel can be formally verified using mathematical methods, meaning that certain dangerous patterns of insulin delivery can be proven impossible, regardless of how complex or opaque the upstream AI controller may be.

For clinical practice, several implications follow. First, it is important to be explicit about whether a system is open loop or closed loop, and what that implies for responsibility. A DailyDose-style app that proposes weekly changes to basal insulin still requires human approval before anything happens; a fully closed-loop system like AIDANET, once activated, can change insulin delivery every few minutes without user input. The expectations, and the counselling needs, are different.

Second, clarity is needed about what is genuinely AI-based and what is not. Many commercial automated insulin delivery systems remain deterministic, even if they are marketed with “smart” or “adaptive” labels. True machine learning appears today in relatively specific places: weekly regimen optimisation in open loop, ML-based meal detection within hybrid closed loop, and in early fully AI-based controllers such as AIDANET. Recognising these differences helps clinicians to anticipate which failure modes to expect and what kind of oversight is realistic.

Third, whenever novel AI-based insulin dosing is considered, the presence of a robust safety layer – conceptually similar to GlucOS – should be part of the evaluation. An AI controller that is not constrained by formal safety checks is difficult to justify in a high-risk domain like insulin dosing. Conversely, a design that separates “smart but fallible” algorithms from a small, provably safe safety kernel offers a more solid foundation for future development.

Insulin dosing is one of the most powerful and dangerous levers in diabetes care. AI has the potential to reduce cognitive burden, personalise therapy and move us closer to fully closed loop, but only within architectures that keep safety, transparency and human oversight at the centre. Understanding where AI already acts today – in open-loop weekly titration, in meal detection, and in prototype controllers like AIDANET – is a crucial step for everyone involved in diabetes technology: clinicians, educators and people living with diabetes alike.

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Lesson 5: AI in Carb Counting

Welcome to this short lesson on AI in carb counting. In the previous lesson on AI in insulin dosing, you saw that some newer systems are beginning to reduce the need

for explicit carbohydrate counting by detecting meals or adapting automatically. But in current practice, most people using insulin for meals still need to estimate carbohydrates, and this remains one of the most error-prone steps in daily diabetes management.

That is why AI-based carbohydrate estimation matters. The key message of this lesson is simple: AI can improve carbohydrate estimation in some settings, especially for simple foods when all ingredients are visible, but current tools are not reliable enough to act as the sole basis for meal bolus decisions. The safest clinical framing is to use AI as a second opinion and educational aid, not as an autonomous dosing tool.

The human baseline

Before judging AI, we first need to know what it is trying to beat. Human carbohydrate counting is often less accurate than patients and clinicians assume. Earlier studies report an average error of about 15 grams per meal [1], with errors rising toward 21 grams in controlled settings [2], which means that even experienced counting is inherently imprecise.

That baseline is important, because an AI tool does not need to be perfect to be useful. It needs to perform at least as well as, and preferably better than, ordinary human estimation. But it also needs to do that consistently, not only on easy foods photographed under ideal conditions.

How these tools work

Most image-based carbohydrate estimation systems follow a three-step pipeline. First, the system identifies the food on the plate, often using a computer vision model such as a convolutional neural network. Second, it estimates portion size, for example through depth inference or 3D reconstruction. Third, it links the result to a nutrition database to estimate carbohydrate content. [3]

The problem is that errors can accumulate across all three steps. If the food is misidentified, or the portion is misjudged, or the nutritional match is wrong, the final carbohydrate estimate can drift substantially. Mixed dishes, sauces, oils, hidden ingredients, and culturally diverse meals make this much harder than a clean image of fruit or bread.

What recent studies show

Several commercial applications have now been evaluated for AI-assisted carbohydrate estimation. One of the best-studied examples is SNAQ, a smartphone application that combines food recognition with portion-size estimation. In a comparative study involving 53 people with type 1 diabetes, SNAQ achieved a mean absolute carbohydrate estimation error of 13.1 grams, outperforming both participant estimates (21 grams) and another commercial application, Calorie Mama (24 grams). [2] These findings suggest that AI-assisted tools can provide meaningful support, particularly for individuals who struggle with carbohydrate counting or whose estimates are highly variable.

The most encouraging recent result comes from a randomized crossover study in adolescents with type 1 diabetes presented at ATTD 2026. In that supervised camp setting, the Carbozen AI photo-recognition app achieved a mean absolute carbohydrate error of 2.1 ± 0.4 grams, compared with 5.3 ± 1.1 grams for traditional counting, and it also reduced estimated insulin dosing error. [4] That suggests AI can outperform standard counting in controlled conditions.

But the picture becomes more mixed outside those ideal settings. In the 2026 study of ChatGPT-4o for image-based carbohydrate estimation, accuracy was high for simple foods such as fruits and vegetables, with about 95 percent agreement, but it dropped sharply for composite meals such as lasagna or stews, where agreement was only 43.3 percent. The study also found that including a size reference object in the image did not always make performance better. [5]

Other 2026 comparisons point in the same direction. [6, 7] General AI tools and diabetes apps can often estimate simple foods equally well. However, these tools still have important limitations. They cannot see details that are not visible in a photo, such as recipe changes, ingredient substitutions, or what is hidden inside layered meals. Home-cooked meals can differ greatly in nutritional content depending on the exact ingredients and preparation methods used. Because of this, photos alone are often not enough, and extra context is needed for accurate estimates.

An important caveat is that many of the currently published studies were performed using earlier generations of large language models. AI systems are evolving rapidly, and newer multimodal models may achieve substantially better performance than the systems evaluated in these studies. As a result, published accuracy figures

should be interpreted as a snapshot of current evidence rather than a fixed limit of what AI-based carbohydrate estimation will be able to achieve in the future.

The safety issue

The most important safety finding is that average performance can look acceptable while rare extreme errors remain dangerous. In a very large ATTD 2026 analysis of more than 456,000 meal descriptions from real type 1 diabetes data, GPT-4.1 and GPT-4.1-nano showed median carbohydrate errors around 6.5 to 7 grams, which is better than the usual human baseline. But the same evaluation also found extreme outlier errors reaching 910 grams and 1,185 grams. [8]

That is the core clinical problem. A median error of 6 or 7 grams may be acceptable for education or cross-checking. A single wildly wrong estimate used to calculate a meal bolus is not acceptable in a high-risk medication context such as insulin dosing. This is why average accuracy alone is the wrong metric for safety-critical use. We must also care about the tail of the error distribution.

Clinical framing

So how should clinicians and educators talk about these tools today? The most practical answer is this: AI carbohydrate estimation can be useful when it supports learning, reflection, and double-checking, especially for simple foods, but it should not be treated as an autonomous carb counter. That is fully consistent with the broader course message that insulin-related decisions require safety layers, human oversight, and clear responsibility.

For patients, this means AI can help as a rough estimator, a training aid, or a second pair of eyes. For clinicians, it means we should ask the same questions as with any AI tool: on what data was it trained, which foods were represented, how was it validated, how large are the outlier errors, and what exactly is the user expected to do with the output?

There is also an equity issue. Current food datasets remain much stronger for standard Western foods than for many South Asian, Middle Eastern, African, or East Asian meal patterns. So for a diverse patient population, underrepresentation is not a theoretical concern but a practical limitation that may directly reduce accuracy.

Finally, it is important to place this in the longer technological direction of travel. AID systems are moving toward reduced dependence on manual carbohydrate entry through meal detection and more automated responses. In that sense, carbohydrate counting is still important today, but it may increasingly become a transitional skill rather than the permanent center of diabetes self-management.

AI in carb counting is therefore promising, but not yet something to trust blindly. Use it to support judgement, not to replace it. In diabetes care, especially when insulin is involved, that distinction remains essential.

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Lesson 6: AI in complication screening

Welcome to this lesson on AI in screening for diabetes complications.

In diabetes care, AI is no longer only about glucose prediction or chatbots. One of the most clinically relevant uses of AI is in helping us detect complications earlier, especially when disease is still silent. In this short lesson, we will use simple visuals to understand where AI already works today, where it is emerging, and why this matters for daily diabetes practice.

Let us begin with the map.

If you look at the screen, you can think of each icon as one complication domain. The eye is where AI is already most established. The kidney, nerve, foot, and heart are the next frontiers.

That distinction is important. Some AI tools are already usable in routine care. Others are promising, but still need more validation and integration into workflow. So the question is not whether AI will be used in complication screening. The real question is where it is mature, and where it is still moving from research into practice.

The retina is the front door of AI screening in diabetes.

This is where the evidence is strongest. A fundus camera takes a retinal image. The image is analyzed by an algorithm. The output is simple and clinically useful: no referable retinopathy, referable retinopathy, or image not gradable.

The reason this matters is visual simplicity. AI turns a specialist image into a triage decision that can be used directly in the diabetes clinic. In other words, screening can move closer to the patient. Instead of relying only on referral to ophthalmology, we can bring screening into the place where the patient already receives diabetes care.

And once you understand that model, you understand the larger future of AI in complications: image in, risk out, action earlier.

Now let us expand that idea.

This slide shows a key concept: one eye photo may tell us more than just retinopathy. This is the idea of oculomics — the eye as a window into systemic disease.

Researchers are exploring whether retinal images can also carry signals of kidney disease, cardiovascular disease, and broader vascular risk. So the same image that we use today to screen for diabetic retinopathy may tomorrow help us identify which patients are at higher renal or cardiovascular risk.

That is a very powerful idea for diabetology, because it means one non-invasive test could provide multiple layers of information. [1]

Let us move to nephropathy.

At present, screening for diabetic kidney disease still relies on the basics: urine albumin and kidney function. And those tests remain essential.

But AI may add a new proactive layer. Some models use retinal images, [2] others use electronic health record data and laboratory patterns [3], to estimate kidney disease risk earlier or more precisely. [4]

So instead of waiting until kidney damage becomes clearly measurable, AI may help us identify who is moving in the wrong direction sooner. The visual on this slide shows that the current gold standard remains urine and eGFR, but AI is emerging as an add-on that may enrich risk stratification.

Next, neuropathy.

Today, neuropathy is often recognized relatively late, and much of screening still depends on symptoms. But symptoms are not an ideal early biomarker.

This is where AI becomes interesting. Here you see two possible digital signals. One comes from corneal confocal microscopy, where AI can analyze tiny nerve-related structures in the cornea. [5] Another comes from short face videos, where subtle vascular patterns may correlate with diabetic neuropathy. [6]

Now, these methods are not yet part of standard routine screening everywhere. But conceptually, they show an important shift: from waiting for pain or complaint, toward using objective digital signals that may reveal neuropathy earlier.

The diabetic foot is another area where AI could become very practical.

This slide is intentionally simple. First, a wound photo is taken. Second, AI analyzes the image. Third, the system flags the level of concern and supports referral.

Why is this useful? Because the diabetic foot often deteriorates between visits, and because visual documentation is already part of good clinical care. If we standardize foot photography and wound follow-up, we create the basis for image-based AI tools that can help classify ulcers, detect infection or ischemia risk, and support earlier intervention.

Again, the value of AI here is not that it replaces expertise. The value is that it may help us react before the foot becomes an emergency.

For many diabetologists, this may be the most important future application: heart failure.

Heart failure is common in diabetes, serious, and often under-recognized, especially in earlier stages when symptoms may be vague or absent.

This slide combines a heart icon with an ECG trace because AI models are increasingly using ECG signals, sometimes combined with EHR data, to detect subclinical cardiac dysfunction. Some reported models in people with diabetes perform better than traditional scores in identifying early left ventricular dysfunction.

Why does that matter? Because if heart failure is recognized earlier, then treatment can start earlier too. And that may mean fewer urgent decompensations, fewer admissions, and better outcomes. [7]

So if you ask where AI may bring the biggest future gain beyond retinopathy, heart failure is a very strong candidate.

AI may also become relevant for earlier detection of ischemic heart disease [8], which is highly important in diabetes because coronary disease often remains silent until late. Beyond ECG and EHR models, AI is increasingly being used in cardiac CT to automate coronary calcium scoring, characterize plaque, and improve risk stratification [9] — even on routine non-contrast, non-ECG-gated chest CTs that were originally acquired for other indications. [10] This is especially interesting in diabetes, where coronary calcium burden strongly refines cardiovascular risk and may help identify patients who need more intensive prevention or further cardiac imaging.

Let us now bring everything back into practice.

This slide is a map for clinicians.

Use now: autonomous AI for retinopathy screening.

Watch closely: retinal AI for kidney and cardiovascular risk.

Prepare data: make sure ECGs, wound images, and structured complication data are well stored and accessible.

Stay realistic: most current tools assist clinical judgement. They do not replace it.

Balance is important. We should neither dismiss AI because it is not perfect, nor overstate it as magic. The real opportunity lies in combining good clinical care with better digital detection.

So let us end with one final idea.

The future of complication screening in diabetes is likely to become earlier, more visual, and more data-driven. Retinopathy screening shows that this is already possible. The next step is to bring the same logic to the kidney, nerves, feet, and heart.

If AI helps us detect complications before symptoms force us to react, then screening becomes more than diagnosis. It becomes an opportunity for earlier protection.

And that is where AI may become truly meaningful in diabetes care.

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Lesson 7: Ethical, Legal, and Privacy Considerations

In this final lesson, we bring everything together: how to use AI in diabetes care in a way that is clinically responsible, legally sound, and organisationally realistic. The core message is simple: when an AI tool suggests something, responsibility does not move to the algorithm. It remains with the human, the clinical team, and the organisation.

That means you should never judge AI only by how intelligent it sounds. You judge it by safety, by fit for your patient population, by privacy, by transparency, and by whether your hospital can implement it properly. Good implementation is not an afterthought. It is a condition for safe use.

1. Use the ETHICS checklist as your first filter

When an AI tool gives an unexpected insulin recommendation, risk score, or screening output, you need a quick clinical framework. The ETHICS checklist is useful for that purpose.

- Equity and fairness: for which patients does the tool work well, and for which patients might it work less well?
- Transparency and patient-centered care: can you explain in plain language what the tool does and why the output should be trusted?
- Human oversight and clinical integrity: can a healthcare professional meaningfully review, correct, and override the output?
- Information privacy and data governance: where does the data go, who has access, and on what legal basis is processing taking place?
- Continuous improvement and sustainability: is there a plan for re-evaluation, update management, and performance monitoring?

- Support and education: are clinicians, educators, and users trained in safe use and in the limitations of the tool?

2. Think actively about bias, privacy, and regulation

AI can widen inequality if training data are too narrow or if, in real-world use, the tool mainly works for digitally confident patients. That is why you should always ask who may be left behind.

Privacy remains a hard boundary. Do not use identifiable patient data in open consumer models without an appropriate agreement, clear legal basis, and local approval. Audio recordings, transcripts, screenshots, and consultation notes are particularly sensitive.

Regulation should not be treated as an afterthought either. With clinical AI, you need to know the intended use, how the tool was validated, how updates are managed, how post-market follow-up is organised, and who is accountable if something goes wrong.

3. If your hospital wants to implement an AI tool, do this

Do not start with the demo. Start with the conditions for safe use.

First, ask for the right documents: what is the exact intended use, in which population was the tool tested, which performance data were reported, what are the known limitations, how does logging work, how are updates managed, and what is the privacy and data-processing framework?

Next, involve the right people: not only enthusiastic physicians, but also diabetes educators, nurses, IT, the privacy lead or DPO, legal, procurement, and if possible, a patient representative.

Then start with a small, clearly defined pilot in one clinical use case, with pre-specified evaluation points such as safety, error patterns, time saved, usability, and workflow impact.

Finally, define explicitly who reviews the output, when AI is advisory only, when a clinician must confirm it, how incidents are reported, and when the implementation should be paused, stopped, or revised.

That is not bureaucracy added next to AI. That is what safe AI implementation is.

4. Non-delegable clinical responsibility

Even with strong or highly automated systems, some responsibilities cannot be delegated. Settings, interpretation of unexpected outputs, communication with patients, and follow-up of incidents remain human responsibilities.

A useful phrase for practice is this: we use AI as a smart assistant and a second opinion, not as a first authority. This mindset keeps you — and your patient — in the driver's seat, even as systems become more capable.